On large language models and human syntax

Timea Szarvas

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The premise

- Why do we model syntactic structure building the way we do?
- I.e. why do we claim that predicates' need to combine with arguments via Merge drives structure building, and that human syntax is not just frequency/probability-based next-word-prediction, informed exclusively by linear structure?

Evidence for hierarchical structure

- Recall what we have learned in session 1:
- Syntactic processes target sensible units that are chunks of a hierarchical structure rather than just linear strings.
- Structural ambiguity arises specifically because there is more to the syntactic structure than the linear string would suggest.
- (Structural ambiguity: whether *in the cabinet* modifies *the cat* or *hiding* in *the mouse that stole the cheese is hiding from the cat in the cabinet* is specified via the hierarchical relation between these constituents. There relations are unrecognizable from the linear string alone.)
- (Phonological processes are also sensitive to constituency boundaries, inducing stress shift if *Chinese dishes* forms a constituent vs. when it does not)
- If syntax would operate on linear structures, these effects would remain mysterious.
- In the theory of syntax you are taught in this class (Minimalism), we represent complex linguistic units as non-linear, hierarchical structures, where the question about which elements form a hierarchical unit (constituency) informs interpretation (LF) and intonation (PF).

Comparing outputs

- One might still say that a LLM is ultimately capable of producing quite human-like sentences, and claim, based on the output of the LMM and human language, that the computational system behind them must be similar.
- It is a logical fallacy that similar or even identical output entails that the path to generating that output must be similar, let alone identical!

⇒ Think of something basic like tying your shoelaces.

- There are (at least??) two ways to do that:
  1. You could tie a knot, then form a loop with each lace and tie the loops together in another knot ⇒ tied shoelaces!

2. Or, you could tie a knot, then form a loop with only one of the laces, wrap the other lace around the base of the loop, and slip it through (at the middle of it) underneath the part of itself wrapped around the original loop (this is really difficult to describe but I guess you understand which method I am trying to get at!) ⇒ also tied shoelaces!

- The output of these two processes will be identical, but the path to the output is different.
- **The output on its own is not a predictor of the path.**
- When presented with tied shoelaces, you have a 50:50 chance of guessing which way they were tied (unless there's yet another method I don't know about).

⇒ Reasoning about the path based on the output is illogical: simply because two systems can generate the same output does not entail that the paths on which they get there have anything in common.

Studying the path leading to the output?

- What's the equivalent to this shoelace example in the debate about similarities between human syntax and LLM performance?
- The important part is not the data they ultimately produce (linguistic structure/tied shoelaces), but the way they get there, and we just established that the equivalence of their output does not entail that they got there via the same operations.
- Since it is impossible to tell what the path to generating structures is (this is why we have a theory about it, because ultimately we do not know) the next best thing we can compare is how humans acquire language vs. how LLMs get trained!

Hallmarks of language acquisition vs. LLM training

- LLMs are fed millions of strings, if not billions or trillions, based on which they compute what sequence of strings is statistically likely vs. unlikely to occur.
- A LLM can only produce what it has encountered before!
- And quite crucially, not just encountered at all, but encountered frequently enough to increase the probability it (the LLM) assigns to the event of encountering it again in a given environment.
- LLMs operate on n-grams (an ordered sequence of strings), not hierarchical structure.
- How do humans acquire language? Not (primarily) based on statistical learning, as the following three examples aim to illustrate.

Want-to-contractions

- Assume that children acquire language in the same manner as LLMs are trained: computing the probability of what they will encounter based on what they have encountered before.
- The following n-gram is encountered frequently in the input of (English-speaking) children:

  (1) Do you wanna play/go/eat/sleep?

- 'Wanna' is the contraction of 'want' and 'to' ⇒ whether the contracted form is allowed depends on whether the linearly adjacent elements 'want' and 'to' are also adjacent in the underlying syntax:[1]

[1] I haven't taught you this bit before, but crucially, in order to get this argument, you need to know that we assume questions like (i) to be derived from sentences that have the basic SVO word order we know from English declaratives. The gap indicates the spot occupied by 'who' in the SVO order.

(2) a. [Who] do you want to/wanna invite __? *object question*
b. [Who] do you want __ to/*wanna win? *subject question*

- Despite the relevant n-gram being identical in (2-a) and (2-b), as well as encountering it all the time in forms like (1), children do not produce 'wanna' when questioning a subject like in (2-b), but do in cases like (2-a) when questioning the object of the clause (Thornton 1990).
- If children would acquire language based on linear strings, their ability to differentiate between the possibility to use 'wanna' in (2-a) and the impossibility of doing so in (2-b) would be entirely unexpected, given that the two strings are identical with respect to their linear structure.

Non-word production

- Similarly, both children and adults are capable of producing and understanding utterances they have never heard before.
- Even when confronted with non-words, humans are able to distinguish whether these non-words were used 'correctly' based on their properties that were implicitly presented (Naigles 1990).
- Suppose we told a human (kid or adult) and an LLM that *gleeb* is an intransitive verb than means "to dance happily".

⇒ The human will reject **Hannah gleebed that Alex left* on the first attempt because of its structural architecture:

⇒ Given that we implicitly provided the information that *gleeb* is a predicate that takes one nominal argument, the human syntax does not know what to do with *that Alex left* as the predicate has no more places for an argument.

- A LLM on the other hand will attempt to find approximates of *gleeb*: based on surface similarity, occurence of similar dance-like words, and words ending in *-eeb*.
- The model will happily estimate the probability of *that Alex left* following *Hannah gleebed* to be non-zero, contrary to humans.
- It will do so precisely because it lacks the hierarchical/relational architecture necessary to instantly rule out such mistakes.

Agreement attraction

- Finally, there's a phenomenon called agreement attraction, where an intervening plural noun phrase "attracts" the verb to agree with it rather than the singular subject that is linearly further away (Bock and Miller 1991 and a plethora of publications since):

  (3) ***The key** to the cabinets **are** rusty.

- Humans can produce such errors — but they stumble over them, realize that they have made a mistake, and correct themselves.
- The error happens due to our working memory (linguistic performance) and is recognized as an error by our knowledge of syntax (linguistic competence)!
- Notice that the ungrammaticality of the sentence does not go away even if we accidentally produce it: this is compatible with the assumption that linguistic competence and performance are separate in humans.

- With a LLM, the probability assigned to encountering a plural verb next increases the more plural noun phrases intervene, and the error in (3) will ultimately become a highly likely outcome:

  (4) *__The key__ to the cabinets in the houses with the gardens cared for by the landlords **are** rusty.

- There is no equivalent to competence in the LLM that marks (4) as ungrammatical: the model operates on frequency and probability; it is insensitive to syntactic rules and hierarchical relations.
- In a sense, the LLM is 'performance-based' all the way down; whereas humans exhibit several traits that cannot be explained without additional machinery.

The bottom line

- Whether one thinks that this is truly the case, i.e. whether additional machinery is necessary to explain human syntax, let alone what this additional machinery should look like, depends entirely on the theory of the human language faculty one adopts.
- The question we started with was frequently asked in recent years, with many opponents of Minimalism claiming that the theory has been refuted given that the output generated by LLMs is so human-like.
- This is a super important conversation, particularly because it offers a great lesson in logic and reasoning, I think (and because, fortunately or unfortunately, many non-linguists will wonder about LLMs and how they relate to linguistics, so as a linguist, it's good to have an idea about what's going on).
- As I have tried to convince you throughout this script, I don't actually think LLMs and the human language faculty have anything in common, or that LLMs offer any valuable insights into human syntax (how would they? and why??).

⇒ Ultimately, both the LLM and human syntax can tie their shoelaces, but how exactly they do it, based on the evidence I presented you with above, seems to be fundamentally different.

- This is crucially what matters: when we study syntactic structures, our primary interest is not what the output looks like (or the fact *that* humans can produce and understand language).

⇒ We are primarily interested in figuring out the path to the output (how the computational system of syntax works) ⇒ this path being the set of rules governing what is and what isn't grammatical, aka linguistic competence, aka human syntax!

⇒ Recall from session 1 that linguistic competence is tacit knowledge, meaning that although humans acquire, possess and use it effortlessly, we are unable to state its rules explicitly.

- This is the whole point about what we are doing: we are building a theoretical model of the language faculty because, given its abstractness, we cannot take a direct look at it.

⇒ We do so by studying the output as exhaustively as possible:

  1. observing it,
  2. testing how specific environments that we have a reasonable understanding of affect it,
  3. hypothesizing what the derivation that generated the output may look like,
  4. testing whether there is evidence for this hypothesis,
  5. and repeating this process until our model can derive the observed empirical facts without overgenerating (so basically until the end of time).

More resources for related perspectives on LLMs

- Bender, Emily M. & Koller, Alexander (2020). Climbing towards NLU: On Meaning, Form, and Understanding in the Age of Data.. *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, p. 5185–5198.
- Katzir, Roni. Large Language Models and human linguistic cognition. GLOWing lecture, June 5th 2024.
- Kodner, Jordan. LLMs and linguistics: Use them for what they are, and don't try to make them what they aren't. 13th Annual Marshall M. Weinberg Symposium "What do Large Language Models Tell Us About Human Language?". Weinberg Institute for Cognitive Science, University of Michigan. Ann Arbor, MI, March 21st 2025.

References

Bock, Kathryn and Carol A. Miller (1991). "Broken agreement". In: *Cognitive Psychology* 23, pp. 45–93.
Naigles, Letitia (1990). "Children use syntax to learn verb meanings". In: *Journal of Child Language* 17.2, pp. 357–374.
Thornton, Rosalind Jean (1990). "Adventures in long-distance moving: The acquisition of complex wh-questions". PhD thesis. University of Connecticut.